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UNIVERSITY OF OSLO

Faculty of Mathematics and Natural Sciences

Mid-term exam IN1900, MAT-IN1105, IN-KJM1900, IN-GEO1900

Date & time: October 10, 2024, 15:00-19:00

Attachments: None

Permitted aids: None. A calculator is available in Inspira

- The exam set consists of 21 multiple choice questions. The maximum score is 25 points.
- All the code in the exam is written in Python 3.
- Most of the questions have one correct answer. A correctly marked answer gives 1 point, wrong or missing answer gives 0 points. There are no negative points for wrong answers, so one should always mark an answer.
- Questions 8, 10, 11 and 15 have 4 answers each. It is given 0.5 points for each correct answer and 0 points for wrong or unmarked answers, maximum 2 points total for each of these questions.

1 Hva skrives ut?

What is printed in the terminal window when the following code is run?

```
a = 7
b = a
a = a + b
print(a, b)
```

Select one alternative:

- ☐ An error message
- ☐ 14 7
- ☐ 7 7
- ☐ 14 14

Maximum marks: 1

2 Hva skrives ut?

What is printed when this code is run?

```
r = 1
t = 2.0
A = r * t
print(type(r), type(A))
```

Select one alternative:

- ☐ <class 'float'>
- ☐ <class 'int'> <class 'float'>
- ☐ <class 'int'> <class 'int'>
- ☐ <class 'float'> <class 'float'>

Maximum marks: 1

3 Hva skrives ut?

What is printed when this code is run?

```
s = 0
for n in range(2, 10, 4):
    s += n
print(s)
```

Select one alternative:

- ☐ 8
- ☐ 18
- ☐ 6
- ☐ An error message

Maximum marks: 1

4 Hva skrives ut?

What is printed in the terminal window when the following code is run?

```
numbers = [10, 25, 50, 40, 70]
result = []
for num in numbers:
    if num < 20:
        result.append("Small")
    elif num < 50:
        result.append("Medium")
    else:
        result.append("Large")

print(result)
```

Select one alternative:

- ☐ ['Small', 'Medium', 'Medium', 'Medium', 'Large']
- ☐ ['Small', 'Medium', 'Large', 'Large', 'Large']
- ☐ ['Small', 'Large', 'Large', 'Medium', 'Large']
- ☐ ['Small', 'Medium', 'Large', 'Medium', 'Large']

Maximum marks: 1

5 Hva skrives ut?

What is printed in the terminal window when this code is run?

```
C = -20
```

```
dC = 5
```

```
Clist = []
```

```
Flist = []
```

```
while C <= 40:
```

```
    Clist.append(C)
```

```
    Flist.append(2 * C + 30)
```

```
    C = C + dC
```

```
print(Clist[-1], Flist[-1])
```

Select one alternative:

- ☐ An error message
- ☐ 40 110
- ☐ 35 100
- ☐ 45 120

Maximum marks: 1

6 Hva skrives ut?

What is printed when this code is run?

```
first_names = ['Minch', 'Han', 'Luke']  
last_names = ['Yoda', 'Solo', 'Skywalker']
```

```
names = [last + first for first, last in zip(first_names, last_names)]  
print(names[-1])
```

Select one alternative:

- ☐ Skywalker
- ☐ An error message
- ☐ LukeSkywalker
- ☐ SkywalkerLuke

Maximum marks: 1

7 Hva skrives ut?

What is printed when this code is run?

```
for i in range(2, 5):
    print(i, end = ' ')
    for j in range(i - 1):
        print(j, end = ' ')
```

The argument **end = ' '** to the print function makes each print end with space (' ') instead of a new line.

Select one alternative:

- ☐ 2 0 1 3 0 1 2 4
- ☐ 2 0 3 0 1 4 0 1 2 5 0 1 2 3
- ☐ 2 0 3 0 1 4 0 1 2
- ☐ 2 3 4 0 0 1 0 1 2

Maximum marks: 1

8 Boolske uttrykk

What are the values of the boolean expressions?

Please match the values:

	False	True
True and (False or True)	☐	☐
not True or False	☐	☐
False or 4 > 3	☐	☐
True and (False and True)	☐	☐

Maximum marks: 2

9 Hva skrives ut?

What is printed when this code is run?

```
hello = "All work and no play makes Jack a dull boy"  
words = hello.split()  
print(type(words), len(words), words[-2])
```

Select one alternative:

- ☐ <class 'list'> 10 dull
- ☐ <class 'str'> 10 boy
- ☐ <class 'list'> dull boy
- ☐ <class 'list'> 10 dull boy

Maximum marks: 1

10 Indeksering i lister

We have the following three lists:

- `kort = [1, 0, -1]`
- `lang = [1, 2, 3, 4, 5]`
- `liste = [1, 2, [1], [2], [[1]], [[2]]]`

To the left of the table are four expressions, and at the top of the table are six possible results when the expressions are evaluated. Choose the right alternative for each expression.

Please match the values:

	1	2	[1]	[2]	[[1]]	[[2]]
<code>liste[-2]</code>	☐	☐	☐	☐	☐	☐
<code>liste[kort[1]]</code>	☐	☐	☐	☐	☐	☐
<code>liste[kort[2] * lang[2]]</code>	☐	☐	☐	☐	☐	☐
<code>liste[lang[kort[0]]]</code>	☐	☐	☐	☐	☐	☐

Maximum marks: 2

11 Lister og Numpy arrays

Assume that this code is run.

```
import numpy as np
dx = 0.02
x = [i * dx for i in range(51)]
y = np.linspace(0, 1, 51)
```

What are the values of the boolean expressions?

Please match the values:

	False	True
$x[-1] \leq 0.5$	☐	☐
$x[50] > 2 * y[50]$	☐	☐
$\text{len}(y) == \text{len}(x)$	☐	☐
$\text{len}(x) == 100$	☐	☐

Maximum marks: 2

12 Hva er riktig?

One of the following statements is correct. Which one?

Select an alternative:

- ☐ A function can only have one return statement.
- ☐ Adding two NumPy arrays of length n will give an array of length $2n$.
- ☐ Adding two lists of length n will result in a list of length n .
- ☐ Adding two NumPy arrays of length n will give an array of length n .

Maximum marks: 1

13 Hva skrives ut?

What is printed when this code is run?

```
import numpy as np
```

```
def sphere(r):  
    return 4 / 3 * np.pi * r**3
```

```
def test_sphere():  
    r = 1.0  
    tol = 1e-8  
    success = abs(sphere(r) - 4 / 3 * np.pi) < tol  
    msg = 'Something wrong in volume'  
    assert success, msg
```

```
test_sphere()
```

Select one alternative:

- ☐ True
- ☐ Nothing is printed
- ☐ Success
- ☐ Something wrong in volume

Maximum marks: 1

14 Hva gjør funksjonen

What does this function do? You can assume that the argument **arg** is a list of strings, for instance ['Norway', 'Sweden', 'Denmark'].

```
def f(arg):  
    m = len(arg[0])  
    for e in arg[1:]:  
        if len(e) > m:  
            m = len(e)  
    return m
```

Select one alternative:

- ☐ The function sorts the list arg from the shortest to the longest string
- ☐ The function returns the longest string in the list
- ☐ The function returns the length of the longest string in the list
- ☐ The function sorts the list arg from the longest to the shortest string

Maximum marks: 1

15 Funksjoner

The code below defines and calls four functions. Some of the functions are correctly defined and used, while others will give an error. Which functions and function calls are correct?

```
def myfun1(x, y, z = 2):
```

```
    return x + y + z
```

```
result = myfun1(2, 3)
```

```
def myfun2(x, y = 3, z):
```

```
    return x + y + z
```

```
result = myfun2(2,3)
```

```
def myfun3(x, y = 3, z = 2):
```

```
    return x + y + z
```

```
result = myfun3(x = 2, z = 1, y = 0.5)
```

```
def myfun4(x, y, z):
```

```
    return x + y + z
```

```
result = myfun4(1, y = 2, 1)
```

Mark the function as correct if both the definition and the function call are correct, and as wrong if one of the two will give an error message.

Mark as correct or wrong

	Wrong	Correct
myfun3	☐	☐
myfun1	☐	☐
myfun4	☐	☐
myfun2	☐	☐

Maximum marks: 2

16 Feilhåndtering

What is printed in the terminal window when the following code is run?

```
import sys
A = [['-1','0','1'],['0','0','0'],['10','9','8']]
try:
    b = int(A[2][-1])
except IndexError:
    print(f'A has length {len(A):d}')
    sys.exit(1)
except TypeError:
    print(f'Cannot convert {A[2]} to int')
    sys.exit(1)
```

`print(b)`

Select one alternative:

- ☐ 10
- ☐ A has length 3
- ☐ 8
- ☐ Cannot convert ['10','9','8'] to int

Maximum marks: 1

17 Hvilken linje mangler?

The file "USD.txt" contains historical exchange rates and has the following structure:

Saturday 21 September 2024 1 USD = 10.484 NOK
Friday 20 September 2024 1 USD = 10.486 NOK
Thursday 19 September 2024 1 USD = 10.499 NOK
Wednesday 18 September 2024 1 USD = 10.584 NOK
Tuesday 17 September 2024 1 USD = 10.605 NOK
Monday 16 September 2024 1 USD = 10.596 NOK
Sunday 15 September 2024 1 USD = 10.645 NOK

We want the code below to read such a file and create two lists, one containing the dates as tuples and one containing the corresponding exchange rates. One line is missing in the code.

```
dates = []  
rates = []
```

```
with open('USD.txt') as infile:  
    for line in infile:  
        words = line.split()  
        dates.append((int(words[1]), words[2], int(words[3])))  
        #missing line goes here
```

Which line must be added for the code to work as intended?

Select one alternative:

- ☐ rates.append(float(words[-2]))
- ☐ rates.append(float(line[-1]))
- ☐ rates = rates + float(words[-1])
- ☐ rates.append(float(words[-1]))

Maximum marks: 1

18 Hvilken linje mangler?

Which line needs to be insterted for the test to pass?

```
def odd_sum(n):  
    # Insert missing line here  
    return sum(s)  
  
def test_odd_sum():  
    expected = 1 + 3 + 5  
    computed = odd_sum(5)  
    msg = f'expected {expected}, got {computed}'  
    assert expected == computed, msg  
  
test_odd_sum()
```

The function **sum**, which is used inside **odd_sum**, is a built-in function in Python. It takes a list, array, or similar object as argument, and returns the sum of its elements.

Select one alternative:

- ☐ s = list(range(0, n+1, 2))
- ☐ s = list(range(1, n+1))
- ☐ s = list(range(n+1, 2))
- ☐ s = list(range(1, n+1, 2))

Maximum marks: 1

19 Hvilket funksjonskall?

The function **Newton(f, df, x, tol = 1e-7)** applies Newton's method to solve an equation on the form $f(x) = 0$. The parameters to the function are the Python function **f** and **df**, which implement the mathematical function $f(x)$ and its derivative, a start value **x** and a tolerance **tol**.

```
def Newton(f, df, x, tol = 1e-7):  
    while abs(f(x)) > tol:  
        x = x - f(x)/df(x)  
    return x, f(x)
```

We want to use the function to solve the equation $x^2 - 2 = 0$. Which function call is correct?

Select one alternative:

- ☐ x0, f = Newton(f(x) = x**2-2, df(x) = 2*x, 2.0)
- ☐ x0, f = Newton(lambda x: x**2-2, lambda x: 2*x, 2.0)
- ☐ x0, f = Newton(f = x**2-2, df = 2*x, 2.0)
- ☐ x0, f = Newton(x**2-2, 2*x, 2.0, 1e-4)

Maximum marks: 1

20 Hva skrives ut?

What is printed in the terminal window when this code is run?

```
def freq_lists(dna_list):
```

```
    n = len(dna_list[0])
```

```
    A = [0]*n
```

```
    T = [0]*n
```

```
    G = [0]*n
```

```
    C = [0]*n
```

```
    for dna in dna_list:
```

```
        index = 0
```

```
        for base in dna:
```

```
            if base == 'A':
```

```
                A[index] += 1
```

```
            elif base == 'C':
```

```
                C[index] += 1
```

```
            elif base == 'G':
```

```
                G[index] += 1
```

```
            elif base == 'T':
```

```
                T[index] += 1
```

```
            index += 1
```

```
    return A, C, G, T
```

```
dna_list = ['TCGCT', 'GGACT', 'GCTGC']
```

```
A, C, G, T = freq_lists(dna_list)
```

```
print(G)
```

Select one alternative:

☐ [0, 0, 5, 0, 0]

☐ 5

☐ [2, 1, 1, 1, 0]

☐ ['G', 'G', 'G', 'G', 0]

Maximum marks: 1

21 Plotting

The Python function `exp_Taylor(N, x)` in the code below approximates the exponential function with a Taylor series with **N** terms. What does the rest of the code do?

```
import numpy as np
import matplotlib.pyplot as plt
from math import factorial

def exp_Taylor(N, x):
    s = np.zeros_like(x)
    for k in range(N+1):
        s += x**k / factorial(k)
    return s

n = 100
x = np.linspace(0, 10, n+1)

for N in range(5,10):
    e = exp_Taylor(N,x)
    plt.plot(x,e,label=f'N={N}')

plt.plot(x, np.exp(x),label = 'Exact')
plt.legend()
plt.show()
```

Select one alternative:

- ☐ Plots the error in the approximation, for N=5 to 10
- ☐ Plots approximations to $\exp(x)$, for N=5 to 9, and the exact function
- ☐ Plots approximations to $\exp(10)$ as a function of N
- ☐ Plots each term in the Taylor series, for N=5 to 9

Maximum marks: 1